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Computer Vision

Final Report

Title:

Street View House Numbers Recognition
Group 57

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Street View House Numbers Recognition

1. Introduction

1.1 Context and Motivation

Automatic house number detection is an important computer vision task with numerous real-world applications, including navigation systems, postal automation, autonomous driving, and smart city infrastructure [2].

However, detecting house numbers in natural scenes is challenging due to variations in illumination, viewing angles, background clutter, and the presence of small or overlapping digits. These challenges make traditional computer vision methods less reliable in complex environments.

Earlier approaches to digit detection relied on handcrafted features such as Histogram of Oriented Gradients (HOG) combined with machine learning classifiers such as Support Vector Machines (SVM) [3,4].

In addition to the YOLO-based detector, a traditional HOG + SVM pipeline was implemented as a baseline method to evaluate the performance difference between classical feature-based approaches and modern deep learning detectors.

Recent advances in deep learning, particularly convolutional neural networks (CNNs), have significantly improved object detection performance. Modern detectors such as Faster R-CNN [5], SSD [6], and the YOLO family [1] can learn hierarchical visual features directly from large-scale datasets.

Among these approaches, the YOLO (You Only Look Once) family of detectors is particularly attractive because it performs object detection in a single forward pass of the network, allowing real-time inference while maintaining competitive accuracy. Therefore, this project adopts a YOLO-based architecture (YOLO26 variant) to develop a robust house number detection system.

1.2 Objectives

The main objectives of this project are:

1. To implement a house number detection system using a YOLO-based object detection framework.
2. To convert the SVHN dataset into YOLO format for digit-level detection.
3. To evaluate different training configurations and analyze their impact on detection performance.
4. To design a post-processing algorithm for reconstructing complete house numbers from detected digits.
5. To implement a baseline digit classifier using HOG features and an SVM classifier for comparison with the YOLO-based approach.

1.3 Expected Outcomes

The expected outcomes of this project include:

- A trained YOLO-based model capable of detecting digits in natural scene images.
- A formatted SVHN dataset compatible with YOLO training.
- A comparative evaluation of multiple training configurations.
- A functional digit grouping module for complete house number reconstruction.
- Quantitative evaluation results including Precision, Recall, and mAP metrics.

1.4 Contributions

The main contributions of this work are:

- Implementation of a complete detection pipeline from data preprocessing to inference.
- Development of a dataset conversion tool from .mat annotation format to YOLO format.
- Comparative analysis of different image resolutions and training strategies.

- Design of a digit grouping algorithm based on spatial alignment and horizontal ordering.
- Implementation and evaluation of a traditional HOG + SVM baseline model for comparison with the YOLO-based detector.

1.5 Report Structure

The remainder of this report is organized as follows:

- Section 2 describes the proposed detection system.
- Section 3 presents the methodology and training strategy.
- Section 4 reports experimental results and analysis.
- Section 5 discusses findings.
- Section 6 concludes the report and suggests future work.

2. Proposed System

2.1 Overall Architecture

The proposed system consists of four main components:

Dataset preparation

Model training

Performance evaluation

Inference and digit reconstruction

The overall workflow is illustrated in Figure 1.

HOUSE NUMBER DETECTION PIPELINE

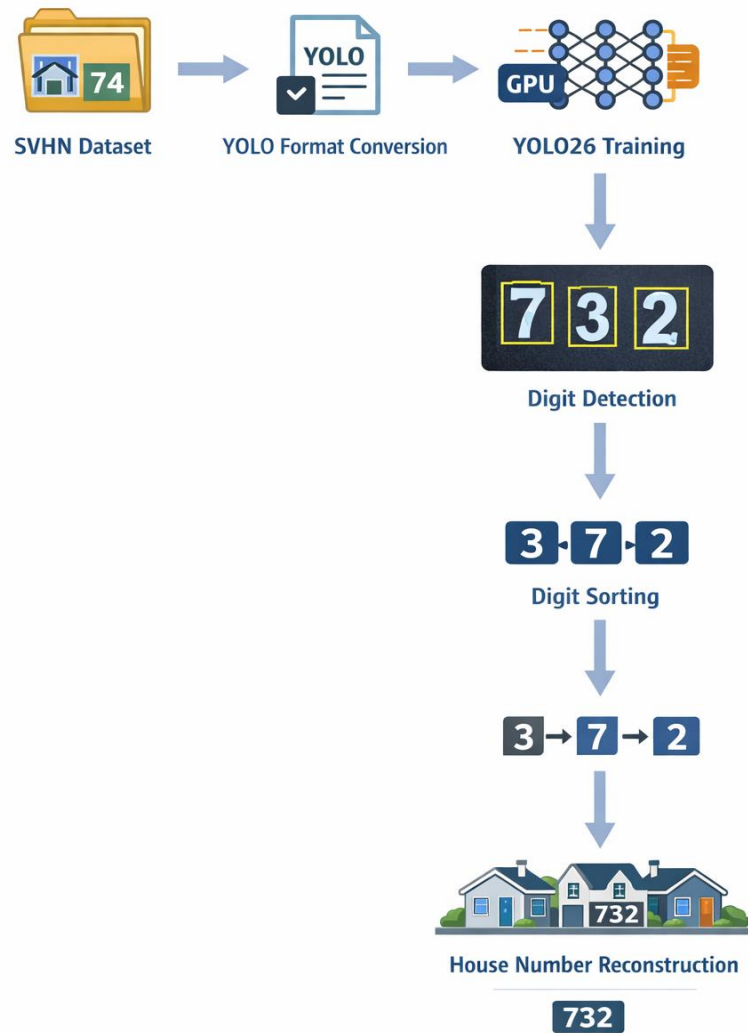


Figure 1: Overall pipeline of the house number detection system

As shown in Figure 1, the proposed system consists of four main stages: dataset preprocessing, model training, performance evaluation, and inference with digit grouping. This pipeline enables the system to detect individual digits and reconstruct complete house numbers.

2.2 Dataset Preparation

2.2.1 SVHN Dataset

The Street View House Numbers (SVHN) dataset contains real-world digit images collected from Google Street View [2]. The dataset includes over 600,000 labeled digit instances extracted from natural scene images and provides bounding box annotations at the digit level.

Compared with MNIST, the SVHN dataset presents several additional challenges, including:

Complex backgrounds

Illumination variations

Digit overlap

Real-world noise

2.2.2 Conversion to YOLO Format

Original annotations are stored in .mat files. A custom preprocessing script was developed to:

Extract bounding box coordinates

Normalize coordinates into YOLO format

Convert class labels

Merge training and extra splits

Use test split as validation

After conversion, all annotations are stored in YOLO format, where each label file contains the class index and normalized bounding box coordinates. This format allows the dataset to be directly used for YOLO-based training.

The final dataset structure is:

```
dataset/  
├── images/  
│   ├── train  
│   └── val  
└── labels/  
    ├── train  
    └── val
```

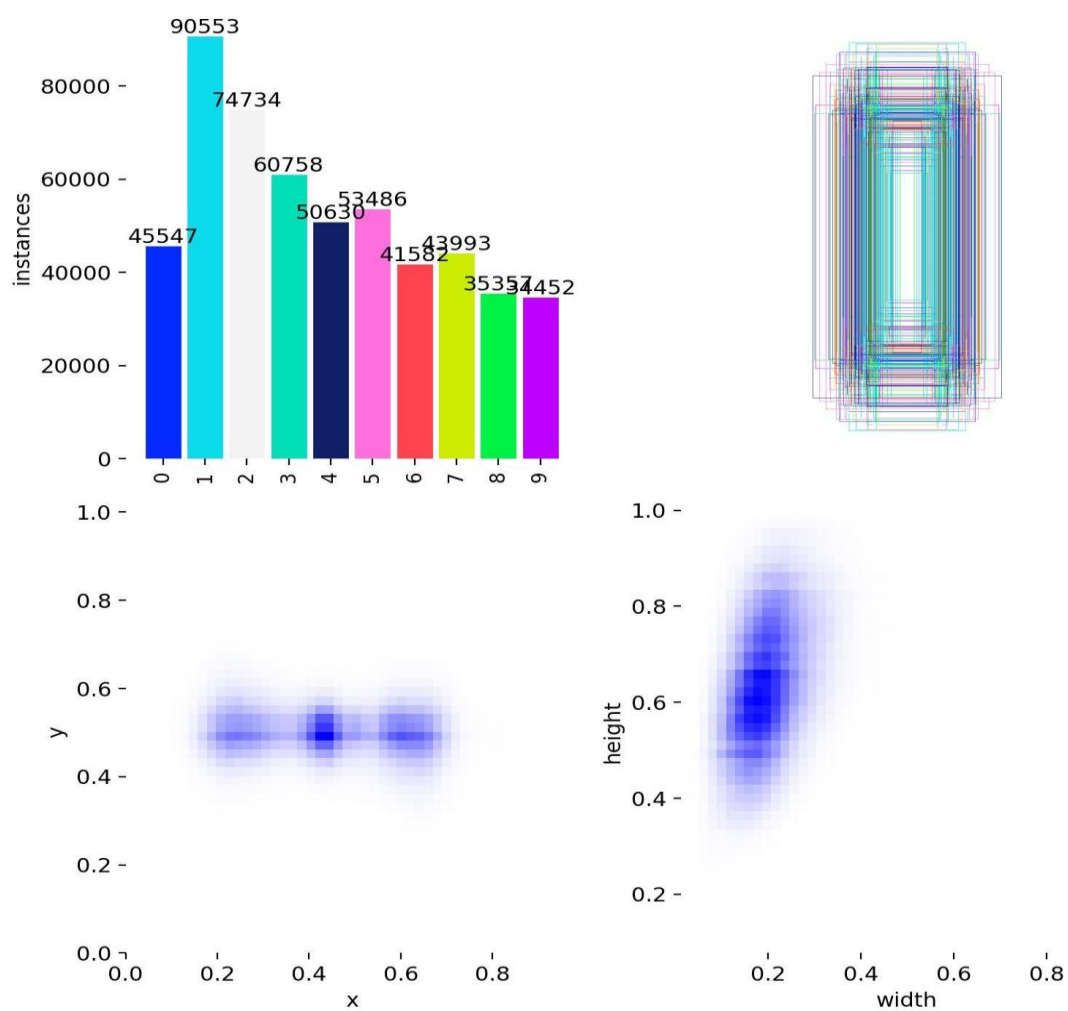


Figure 2: Digit class distribution and bounding box density.

2.3 Detection Pipeline

During inference, the following steps are applied:

1. Detect individual digits using the trained YOLO model.
2. The detected bounding boxes are then sorted by their x-coordinate so that digits are ordered from left to right, which corresponds to the natural reading order of house numbers.
3. Group digits based on horizontal proximity and vertical alignment.
4. Concatenate sorted digits to reconstruct the full house number.

An example output produced by the digit grouping algorithm is shown in Listing 1. The system first detects individual digits using YOLO and then groups them based on spatial proximity and vertical alignment to reconstruct full house numbers

This grouping strategy assumes that digits belonging to the same house number are horizontally aligned and appear close to each other in the image. Therefore, digits are grouped by comparing their spatial distances and vertical alignment.

Example detection results are shown in Figure 3.

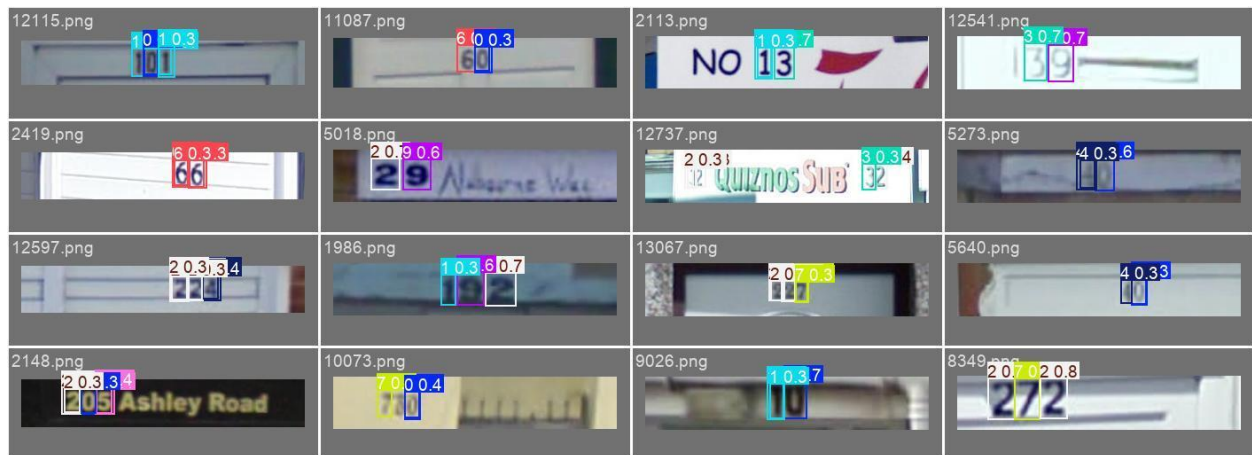


Figure 3: Multi-digit grouping.

2.4 Related Work

Object detection has been extensively studied in computer vision. Early deep learning-based detectors can generally be divided into two categories: two-stage detectors and single-stage detectors.

Two-stage detectors such as Faster R-CNN first generate region proposals and then classify each region. These approaches usually achieve high detection accuracy but require more computational resources and inference time.

Single-stage detectors, including SSD [6] and YOLO [1], perform detection and classification simultaneously in a single network. This design significantly improves inference speed and makes these models suitable for real-time applications.

For house number recognition, the Street View House Numbers (SVHN) dataset has become a widely used benchmark for evaluating digit detection systems. Many recent studies apply convolutional neural networks to detect and classify digits directly from natural images.

In this project, a YOLO-based detector is adopted due to its strong balance between detection accuracy and real-time performance. This design makes YOLO particularly suitable for applications that require fast and accurate detection in real-world environments.

3. Methodology

3.1 YOLO Architecture

YOLO (You Only Look Once) is a single-stage object detection framework that performs object localization and classification simultaneously. Instead of generating region proposals like two-stage detectors, YOLO formulates object detection as a single regression problem that predicts bounding boxes and class probabilities simultaneously.

The input image is divided into a grid, and each grid cell predicts bounding boxes together with confidence scores and class probabilities. During training, the network learns to minimize both localization and classification errors.

During inference, Non-Maximum Suppression (NMS) is applied to remove overlapping bounding boxes and retain the most confident predictions.

This unified detection framework allows YOLO to achieve high inference speed compared with traditional multi-stage detectors.

3.2 HOG + SVM Baseline Method

Example detection results produced by the HOG + SVM baseline model

- (a) threshold = 0.45



- (b) threshold = 0.40



- (c) threshold = 0.35



Figure 4: Example detection results produced by the HOG + SVM

To provide a baseline comparison, a traditional digit detection pipeline based on Histogram of Oriented Gradients (HOG) features and a Support Vector Machine (SVM) classifier was implemented.

The figures illustrate example detection results produced by the HOG + SVM model on several street-view images containing house numbers. The detector is able to identify several digits correctly, such as “1680”, “736”, “426”, and “209”.

However, several limitations can be observed. In some cases, background patterns such as decorations or architectural structures are incorrectly classified as digits, resulting in false positive detections. Additionally, the detector occasionally fails to detect digits with low contrast or unusual shapes.

The Histogram of Oriented Gradients (HOG) descriptor is used to extract gradient-based features from digit images. These features capture the local edge orientations which are effective for representing digit shapes.

3.3 Training Configuration

Two configurations were evaluated:

Table 1: Training configurations used in the experiments.

Configuration	Image Size	Batch	Epochs
A	640	32	10
B	320	128	30

The model was trained using stochastic gradient descent (SGD) with a learning rate of 0.01 and momentum of 0.937. Data augmentation techniques including Mosaic augmentation and random scaling were applied to improve model generalization.

3.4 Evaluation Metrics

The following metrics were used:

- Precision
- Recall
- mAP@0.5
- mAP@0.5:0.95
- F1-score

Precision measures the proportion of correctly predicted detections among all predicted detections. Recall measures the proportion of correctly detected objects among all ground truth objects.

Mean Average Precision (mAP) is used to evaluate detection accuracy by computing the area under the precision–recall curve. The mAP@0.5 metric uses an Intersection over Union (IoU) threshold of 0.5, while mAP@0.5:0.95 averages the performance across multiple IoU thresholds.

4. Experiments and Results

The experimental results indicate that the optimized configuration achieves the best trade-off between detection accuracy and computational efficiency. Although higher resolution provides more spatial details, increasing the number of training epochs allows the model to converge better and improves detection performance.

Furthermore, the SVHN dataset mainly contains digits with relatively large visual sizes, which means that reducing the image resolution does not significantly affect detection performance.

The per-class performance analysis indicates that most digits achieve high precision and recall. However, digit ‘1’ shows slightly lower recall due to its narrow shape and visual similarity with background structures.

Overall, the results confirm that the proposed YOLO-based approach is effective for house number detection in real-world images. One possible explanation is that additional training epochs allow the model to better learn the visual variations

present in the SVHN dataset. Furthermore, smaller input resolution may encourage the network to focus on more robust features instead of relying on fine-grained pixel details.

These observations suggest that careful tuning of training parameters is essential for achieving optimal performance in digit detection tasks.

4.1 Experimental Setup

Training was conducted on a system with:

- Intel i5-14600K CPU
- RTX 3060 (12GB GPU)
- 32GB RAM

The experiments were implemented using the Ultralytics YOLO framework built on top of the PyTorch deep learning library.

Each experiment was performed once with consistent hyperparameters except for image size and batch size.

Example input images from the SVHN dataset used for evaluation are shown in Figure 4.

These images contain multiple digits representing house numbers captured in real-world environments.



Figure 5: Example input images containing house numbers from the SVHN dataset.

4.2 Comparative Performance

Table 2 presents the comparison between the two configurations.

Table 2: Performance comparison between training configurations.

Setting	mAP@0.5	mAP@0.5:0.95	Training Time
640×10 epochs	0.890	0.425	~10h
320×30 epochs	0.904	0.451	~8h

The optimized configuration achieved better overall performance. The results show that increasing the number of training epochs has a greater impact on performance than increasing image resolution. Although higher resolution provides more spatial details, the model benefits more from longer training.

The precision–recall relationship of the trained detector is illustrated in Figure 5.

The curve shows the trade-off between precision and recall across different confidence thresholds and provides insight into the detector’s overall performance.

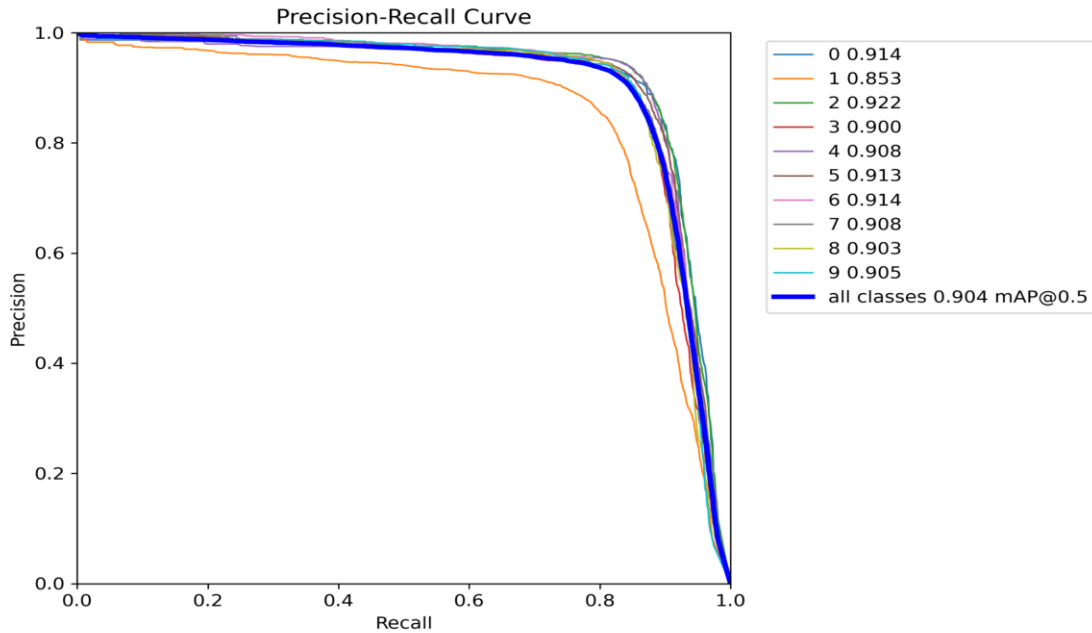


Figure 6: Precision–Recall curve of the trained YOLO detector on the validation dataset.

4.3 Discussion on Training Configuration

Several factors contributed to the improved performance of the 320×320 configuration:

- More training epochs allowed better convergence.
- Larger batch size stabilized gradient updates.
- SVHN digit size is sufficiently large for 320 resolution.
- Longer training compensated for reduced spatial resolution.

4.4 Per-Class Performance

Table 3 shows per-digit performance metrics.

Table 3: Per-class detection performance on the SVHN dataset.

Digit	Images	Instances	Precision	Recall	mAP@0.5	mAP@0.5:0.95
0	1,687	1,744	0.899	0.872	0.914	0.463
1	4,578	5,099	0.891	0.762	0.853	0.369
2	3,833	4,148	0.918	0.864	0.922	0.468
3	2,696	2,882	0.910	0.840	0.900	0.454
4	2,397	2,523	0.912	0.868	0.908	0.442
5	2,289	2,383	0.902	0.861	0.913	0.469
6	1,922	1,977	0.899	0.849	0.914	0.468
7	1,943	2,019	0.916	0.842	0.908	0.451
8	1,612	1,660	0.910	0.846	0.903	0.457
9	1,555	1,595	0.912	0.847	0.905	0.472

Table 3 presents the per-class detection performance for each digit in the SVHN dataset. Most digits achieve high precision and recall values, indicating that the model is able to accurately detect digits in natural scene images.

Among all digits, digit ‘1’ shows the lowest recall (0.762) and the lowest mAP@0.5 (0.853). This is mainly due to its narrow shape and visual similarity with background structures such as vertical edges and poles, which makes it more difficult for the detector to localize correctly.

Digits such as “2”, “5”, and “6” achieve the highest detection performance with mAP@0.5 values above 0.91, suggesting that these digits have more distinctive visual patterns that are easier for the model to learn. In addition, class imbalance in

the dataset may also influence detection performance, since some digits appear more frequently than others in the SVHN dataset.

Overall, the results indicate that the proposed YOLO-based detector performs consistently across most digit classes.

Figure 7 illustrates qualitative detection results of the trained YOLO detector. The bounding boxes highlight detected digits along with their predicted class labels and confidence scores.

These examples demonstrate the ability of the model to detect multiple digits and reconstruct complete house numbers in real-world images.



Figure 7: Example detection results produced by the YOLO-based house number detector.

To further analyze classification errors between digit classes, the confusion matrix of the trained model is shown in Figure 8.

The confusion matrix highlights which digits are most frequently misclassified and helps identify patterns of confusion between visually similar digits.

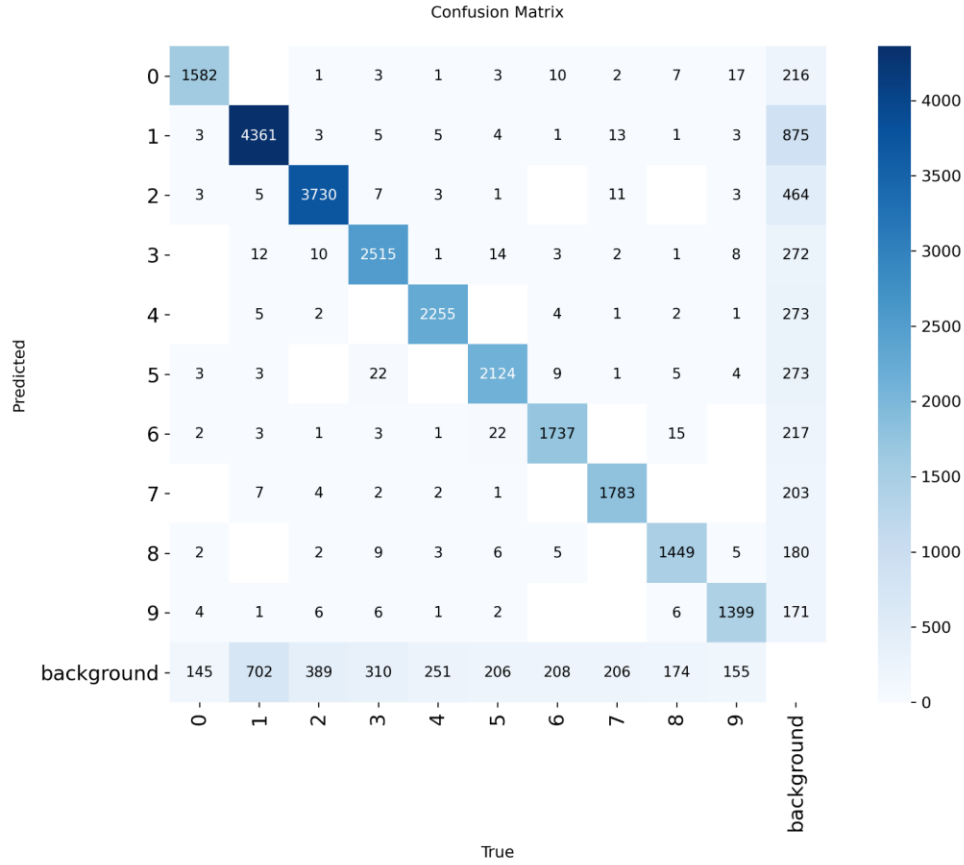


Figure 8: Confusion matrix of digit detection results on the validation dataset.

4.5 Inference Speed

Table 4 summarizes the inference speed of the proposed system.

Table 4: Inference speed of the proposed detection pipeline

Stage	Time
Preprocessing	0.0 ms
Inference	0.4 ms
Postprocessing	0.1 ms

An example inference log produced by the detection system is shown below.

Input image resolution: 544×1280

Speed: 4.2 ms preprocessing, 47.7 ms inference, 1.5 ms postprocessing per image

The proposed system demonstrates real-time detection capability. Such inference speed makes the system suitable for practical applications such as navigation systems and real-time visual recognition. The inference time was measured on an

NVIDIA RTX 3060 GPU, demonstrating that the system can perform real-time digit detection.

4.6 HOG + SVM Baseline Performance

The HOG + SVM baseline model achieved an overall accuracy of 0.86 on the SVHN test dataset.

The macro-averaged precision, recall, and F1-score are 0.84, 0.84, and 0.84 respectively.

Although the baseline model can correctly classify many digits, it tends to produce false positives on background structures and struggles with digits that have similar visual patterns.

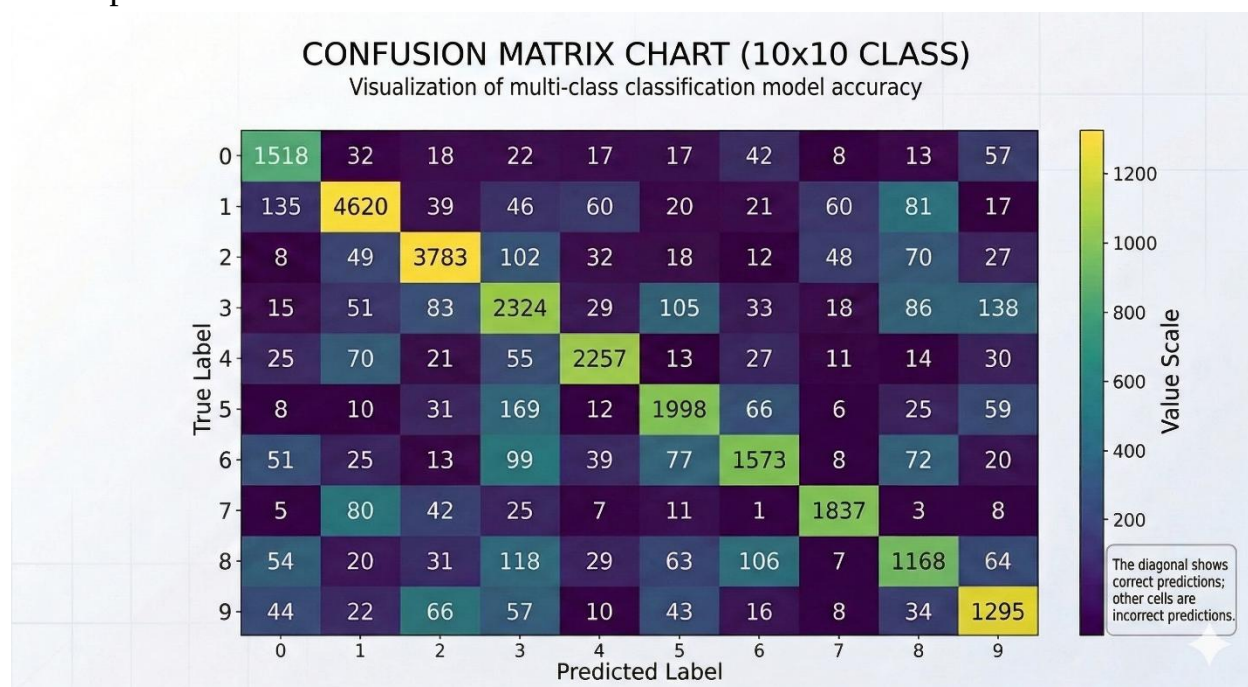


Figure 9: Confusion matrix of the HOG + SVM digit classifier

4.7 Parameter Sensitivity Analysis

To better understand the behavior of the proposed detection pipeline, additional experiments were conducted to analyze the influence of several inference parameters, including the confidence threshold, the Intersection-over-Union (IoU) threshold used in Non-Maximum Suppression (NMS), and the input image resolution.

First, different confidence thresholds were evaluated to analyze the trade-off between precision and recall. A high confidence threshold reduces false positives

by keeping only predictions with high confidence scores, but it may also miss digits that have moderate confidence values. On the other hand, a very low threshold increases recall but introduces more false detections from background structures. Based on the experimental observations, a confidence threshold of **0.3** provides the best balance between detecting most digits while keeping the number of false positives relatively low. Therefore, this value was selected for the final detection pipeline.

Second, the influence of the IoU threshold used in the NMS stage was analyzed. IoU values ranging from **0.2 to 0.6** were tested. The experimental results show that the detection outputs remain almost identical across all tested IoU values. This behavior occurs because digits in the SVHN images rarely overlap significantly. As a result, the NMS step removes very few bounding boxes regardless of the IoU threshold. Therefore, the IoU parameter has minimal impact on the final detection results for this task. An IoU threshold of **0.2** was selected for consistency in the final system.

Finally, the effect of input image resolution was investigated using image sizes of **320, 640, 960, and 1280 pixels**. The results indicate that lower resolutions such as 320 and 640 may lead to missed detections because small digits lose spatial details when the image is resized. Increasing the resolution improves the model's ability to detect small digits. In particular, resolutions of **960 and 1280** produce more complete detection results, successfully identifying all digits in the scene. Although higher resolutions slightly increase computational cost, they significantly improve detection reliability. Therefore, a higher resolution (**960–1280 pixels**) was adopted for the final inference pipeline.

Overall, these experiments show that the detection system is relatively insensitive to the IoU threshold but strongly influenced by the confidence threshold and the input image resolution. The final inference configuration used in this work is summarized as follows:

- **Confidence threshold:** 0.3
- **IoU threshold:** 0.2
- **Input image size:** 960–1280

This configuration ensures that most digits are detected correctly while maintaining stable detection performance in complex street-view images.

Table 5: Selected parameters for the final inference pipeline.

Parameter	Tested Values	Selected Value
Confidence threshold	0.1 – 0.5	0.3
IoU threshold	0.2 – 0.6	0.2
Image size	320 – 1280	960 / 1280

5. Discussion

The experimental results demonstrate that the YOLO-based approach achieves high detection performance on real-world digit images. The model generalizes well across varying lighting conditions and backgrounds.

The post-processing grouping algorithm successfully reconstructs multi-digit house numbers with minimal errors.

Despite promising results, several limitations remain. One common failure case observed during evaluation is missed detection. In some situations, the detector fails to detect certain digits even though they are clearly visible in the image. This typically occurs when digits are partially occluded, appear very small, or have low contrast with the background.

Compared with the HOG + SVM baseline model, the YOLO-based detector provides higher detection accuracy and better robustness to complex backgrounds. Figure 10 illustrates an example where one or more digits in a house number are not detected by the model.



Figure 10: Example of missed detection cases where the model fails to detect one or more digits in a house number.

The detector may occasionally misclassify visually similar digits such as 4 and 6, and the recall for digit ‘1’ is lower due to its thin structure and similarity to background edges. Future improvements may include using more advanced YOLO variants, applying stronger data augmentation techniques, or integrating sequence modeling approaches to improve multi-digit recognition. These findings suggest that both model architecture and dataset characteristics play important roles in determining detection performance.

6. Conclusion and Future Work

This report presented a YOLO-based approach for house number detection using the SVHN dataset. The system includes data preprocessing, model training, performance evaluation, and a digit grouping module for reconstructing complete house numbers.

Experimental results demonstrate that the proposed pipeline is capable of detecting digits in natural scene images with competitive accuracy. The study also highlights the importance of training configuration in influencing detection performance.

Experimental comparisons show that the YOLO-based detector outperforms traditional feature-based approaches such as HOG + SVM in both detection accuracy and robustness.

Future work may explore improved detection architectures, larger training datasets, and more advanced post-processing techniques for multi-digit recognition.

The system achieves:

- Precision: 0.907
- Recall: 0.845
- mAP@0.5: 0.904
- mAP@0.5:0.95: 0.451

Future work may include the following directions:

- Training longer at higher resolutions
- Using larger YOLO variants
- Applying multi-scale inference
- Deploying the model for real-time edge applications

Overall, the results demonstrate that YOLO-based detectors provide an effective solution for real-world house number detection tasks. These observations highlight the importance of both model architecture and training configuration in achieving reliable digit detection performance.

7. References

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